

This SKY130 Design Rule Manual documents additions and updates to the SKY130 Open-Source DRM and DRC files. This manual is based on:

- Open-Source SKY130 PDK documentation version “SkyWater SKY130 PDK 0.0.0-369-g7198cf6”.
- Information provided by SkyWater Technology.

Designations

CDR : Critical Design Rule. Violations of these rules cannot be waived.

CDRW: Critical Design Rule, but *possibly* Waivable; SkyWater Technology must approve any waivers.

GUIDELINE: Rule is only a guideline.

WARNING: Rule is only a warning.

Note on “CORE” regions

The layer “areaid.ce” (GDSII layer 81:2 and Virtuoso DFII layer areaid:core) is used to designate core regions. These areas have less stringent design rules. Some rules are applied only outside of areaid.ce only and do not necessarily have a corresponding “in core” check. The presence of areaid.ce shapes will flag rule x.8. Use of the areaid.ce layer must be approved by SkyWater Technology.

Note on mcon rules

To be consistent with SkyWater libraries, files and documentation, rules previously designated “ct” rules are renamed “mcon.” Examples: mcon.1 / mcon.2 / mcon.cover.1 / mcon.cover.2

Copper (CU rules)

CU option is not currently offered for this PDK; rules are retained for reference.

RDL Option

RDL is not currently offered for this PDK; rules have been removed.

SRAM Alternate Rules

Alternate rules for SRAM regions are enabled with environment variable / deck switch “SRAM” – this is for debugging purposes only and will cause real licon.4a / licon.4b / mcon.cover.1 violations to be missed. The purpose of this switch is to allow users to more easily find violations that are outside of SRAM areas.

Switches

The following switches are defined in the DRC files:

Turn On Copper Rules (Off by default): CU

Turn Off Density Rules (On by default): NODEN

Turn Off Guideline Rules (On by default): NOGLR

Use Alternate SRAM Rules (Off by default): SRAM

Turn Off Warning Rules (On by default): NOWARN

5.7.1 General FIXME rules (x.-)

Rule	Description	Value
x.1a	p1m.md (OPC), DECA, and AMKOR layes (pi1.dg, pmm.dg, rdl.dg, pi2.dg, ubm,dg, bump.dg) and mask data for for p1m, met1, via1, met2 must be on a grid of: <i>This rule not implemented</i>	0.001 um
x.1b	Except for layers mentioned in 1a, data for SKY130 layout and mask on all layers must be on a grid of: Exempt for shapes inside Seal Ring (not currently excluded)	0.005 um
x.2 Rules	<i>Shapes on layers listed by Rule x.2 are only permitted angles:</i>	90°
x.2.licon	CDR: licon shapes must be rectangles	
x.2.mcon	CDR: mcon shapes must be rectangles	
x.2.via1	CDR: via1 shapes must be rectangles	
x.2.via2	CDR: via2 shapes must be rectangles	
x.2.via3	CDR: via3 shapes must be rectangles	
x.2.via4	CDR: via4 shapes must be rectangles	
x.2a	Analog circuits identified by areaid.analog to use rectangular diff and tap geometries only, not to be merged into more complex shapes (T's or L's) <i>This rule not implemented</i>	
x.3 Rules	<i>Angles permitted on all other layers except WLCSP layers:</i>	45°
x.3.nwell	nwell has non-45 degree edge	
x.3.dnwell	dnwell has non-45 degree edge	
x.3.hvtp	hvtp has non-45 degree edge	
x.3.diff	diff has non-45 degree edge	
x.3.tunm	tunm has non-45 degree edge	
x.3.poly	poly has non-45 degree edge	
x.3.npc	npc has non-45 degree edge	
x.3.nsdms	nsdms has non-45 degree edge	
x.3.psdms	psdms has non-45 degree edge	
x.3.li1	li has non-45 degree edge	
x.3.m1	met1 has non-45 degree edge	
x.3.m2	met2 has non-45 degree edge	
x.3.m3	met3 has non-45 degree edge	
x.3.m4	met4 has non-45 degree edge	
x.3.m5	met5 has non-45 degree edge	
x.3.nsm	nsm has non-45 degree edge	
x.3.pad	pad has non-45 degree edge	
x.3.hvntm	hvntm has non-45 degree edge	
x.3.capacitor	capacitor has non-45 degree edge	
x.3.ncm	ncm has non-45 degree edge	
x.3.inductor	inductor has non-45 degree edge	
x.3.hvtr	hvtr has non-45 degree edge	
x.4.EM	Electromigration Guidelines are not available; not covered by DRC	
x.5.nwell	All pin polygons must be within the 'drawing' polygons of the layer - nwell	
x.5.diff	All pin polygons must be within the 'drawing' polygons of the layer - diff	
x.5.poly	All pin polygons must be within the 'drawing' polygons of the layer - poly	
x.5.li	All pin polygons must be within the 'drawing' polygons of the layer - LI	
x.5.m1	All pin polygons must be within the 'drawing' polygons of the layer - met1	
x.5.m2	All pin polygons must be within the 'drawing' polygons of the layer - met2	
x.5.m3	All pin polygons must be within the 'drawing' polygons of the layer - met3	
x.5.m4	All pin polygons must be within the 'drawing' polygons of the layer - met4	
x.5.m5	All pin polygons must be within the 'drawing' polygons of the layer - met5	

x.5.pad	All pin polygons must be within the 'drawing' polygons of the layer – pad	
x.6	All intra-layer separation checks will include a notch check; <i>This implemented for all self-layer spacing checks</i>	
x.7	Mask layer line and space checks must be done on all layers	
x.8	CDRW: WARNING: Use of areaid.ce layer (memory core ID layer) must be approved by SkyWater Technology; some rules apply only to areas outside of areaid.ce and a core rule may not exist.	
x.9	Shapes on maskAdd or maskDrop layers (serifs) are allowed in core only <i>Rule not implemented</i>	
x.10.diff	res purpose layer for diff cannot overlap licon	
x.10.poly	res purpose layer for poly cannot overlap licon	
x.11	Fuses are no longer supported	
x.n12ar	To comply with the minimum spacing requirement for layers in the frame: - Min spacing of areaid.mt to any non-ID layer - Min enclosure of any non-ID layer by areaid.mt - Rules exempted for cells with name “*_buildspace” <i>These rules not implemented due to lack of information</i>	NV
x.12d	Min spacing of areaid.mt to huge_metX (met3 & met5 exempt) <i>This rule not implemented</i>	NV
x.12e	Min enclosure of huge_metX by areaid.met (met3 & met5 exempt) <i>This rule not implemented</i>	NV
x.13	Min spacing between features located across areaid.ce <i>This rule not implemented</i>	NV
x.14	Min width of features straddling areaid.ce <i>This rule not implemented</i>	NV
x.15a	Drawn compatible, mask, and waffle-drop layers are allowed only inside areaid:mt (i.e., etest modules), or inside areaid:sl (i.e., between the outer and inner areaid:sl edges, but not in the die) or inside areaid:ft (i.e., frame, blankings). Exception: FOM/P1M/Metal waffle drop are allowed inside the die <i>This rule not implemented</i>	
x.15b	Rule X.15a exempted for cpmmm.dg inside cellnames “PadPLfp”, “padPLhp”, “padPLstg” and “padPLwlb” (for the SKY130di-5r-gsmc flow) <i>This rule not implemented</i>	
x.16	Die must not overlap areaid.mt (rule waived for test chips and exempted for cellnames “tech_CD_”, “_techCD_”, “lazX_” or “lazY_”) <i>This rule not implemented</i>	
x.17	All labels must be within the “drawing” polygons of the layer <i>This rule not implemented</i>	
x.18	Use redundant mcon, via, via2, via3 and via4. Single contacts and vias under areaid.ce and areaid.standarc are excluded from the single via check. <i>This rule not implemented</i>	
x.19	Lower left corner of the Seal Ring should be at origin (0,0)	
x.20	Min spacing between pins on the same layer (center to center) <i>Rule not implemented due to lack of information</i>	NV
x.21	prunde.dg is allowed only inside areaid.mt or areaid.sc <i>Rule is obsolete</i>	
x.22	No floating interconnects (poly, li1, met1-met5) or capm allowed; rule flags interconnects with no path to poly, diff tap or metal pins. Only metals outside areaid.stdcell are checked. The following are exempt from x.22 violations: “*_techCD_top*” cells, inductor.dg, moduleCut, capacitors <i>Recommended only; rule not implemented</i>	
x.23a	<i>Rule is obsolete</i>	

x.23b	<i>Rule is obsolete</i>	
x.23c.poly	areaid:seal (Seal Ring) must not overlap poly	
x.23c.li1	areaid:seal (Seal Ring) must not overlap li1	
x.23c.m1	areaid:seal (Seal Ring) must not overlap met1	
x.23c.m2	areaid:seal (Seal Ring) must not overlap met2	
x.23c.m3	areaid:seal (Seal Ring) must not overlap met3	
x.23c.m4	areaid:seal (Seal Ring) must not overlap met4	
x.23c.m5	areaid:seal (Seal Ring) must not overlap met5	
x.24	<i>Rule is obsolete</i>	
x.25	<i>Rule is obsolete</i>	
x.26	<i>Rule is obsolete</i>	
x.27	<i>Rule is obsolete</i>	
x.28	Min width of areaid.sl <i>Rule not implemented due to lack of information</i>	NV
x.29	<i>Rule is obsolete</i>	
x.fuse	CDRW: WARNING: The met4 fuse layer (GDS 71:17) is no longer supported	
X.moduleCut	CDRW: WARNING: Use of the areaid.mt (moduleCut) layer (GDS 81:10) use is SkyWater Technology only	
x.pad	WARNING: pad pcells should be used for bondpad Rule not implemented	
x.photo	CDRW: WARNING: Use of areaid:photo layer (GDS 81:81) is prohibited	
x.prBndry	CDRW: WARNING: Use of prBoundary layer (GDS 235:4) is prohibited	
x.pwell	CDRW: WARNING: pwell must connect by ptap at least once	
x.RDL	CDRW: WARNING: Use of the RDL layer (GDS 74:20) is prohibited	
x.target	CDRW: WARNING: Use of the fuse target layer (GDS 76:44) is prohibited	
x.tunm	CDRW: WARNING: Use of the tunm layer (GDS 80:20) is prohibited	

5.7.2 dnwell rules

Rule	Description	Value
dnwell.2	CDRW: Min width of deep nwell	3.0 um
dnwell.3	CDRW: Min spacing/notch between deep nwells. exempt inside UHVI	6.3 um
dnwell.3a	Min spacing of dwnell in v20 on same net	2.5 um
dnwell.3d	Min spacing of dwnell in v20 not on same net	12. um
dnwell.3b	Min spacing of dnwell in uhvi to dnwell outside uhvi	12. um
dnwell.3c	Min spacing of dnwell in v20 to nwell outside v20	9.5 um
dnwell.4	Dnwell can not overlap pnp	
dnwell.5	P+ Diff can not straddle dnwell	
dnwell.6	RF NMOS must be enclosed by deep nwell	
dnwell.7	Dnwell can not straddle areaid: substratecut	

5.7.3 nwell rules

nwell.1	CD: Min width of nwell	0.84 um
nwell.2a.lv	CDR: Min space/notch between two LV n-wells Manual merge wells if less than minimum	1.27 um
nwell.2a.lv_hv	Min space of LV_nwell to HV_nwell:	1.27 um
nwell.2a.lv_vhv	Min space of LV_nwell to VHV_nwell:	2.0 um
nwell.2a.lv_uhv	Min space of LV_nwell to UHV_nwell:	2.5 um
nwell.4	All n-wells will contain metal-contacted tap (checks only for licon on tap). exempted from high voltage cells inside UHVI. NOTE: exceptions have NOT been considered, there may be false errors	

nwell.5	Deep nwell must be enclosed by nwell by at least Exempted inside UHVI (layer v20) or areaid.lw Nwells can merge over deep nwell if spacing too small NOTE: exceptions have NOT been considered, there may be false errors NOTE: areaid.lw exemption not coded, areaid.lw not in PDK or DRM	0.4 um
nwell.5a	min enclosure of nwell by dnwell inside UHVI No value in DRM, not coded	NV
nwell.5b	nwell inside UHVI must not be on the same net as nwell outside UHVI	
nwell.6	Min enclosure of nwell hole by deep nwell outside UHVI shapes in cell *_tech_CD_top exempted	1.03 um
nwell.7	Min spacing between nwell and deep nwell on separate nets	4.5 um
	Spacing between nwell and deep nwell on the same net is set by the sum of the rules nwell.2 and nwell.5. By default, DRC run on a cell checks for the separate-net spacing, when nwell and deep nwell nets are separate within the cell hierarchy and are joined in the upper hierarchy. In this case, the joinNets switch should be turned on. waffle_chip shapes in cell *_tech_CD_top exempted. TBA	NV
nwell.ntap	nwell must have at least one ntap	
nwell.sameNet	WARNING: Different voltage nwell (LV, hvi, vhvi, uhvi) found on the same net	

5.7.4 pwbm Rules

Note: pwbm rules are not implemented due to lack of information.

pwbm.1	Min width of pwbm	NV
pwbm.2	Min spacing between pwbm inside UHVI	NV
pwbm.3	Min enclosure of dnwell by pwbm inside UHVI (exempt pwbm hole inside dnwell)	NV
pwbm.4	dnwell inside UHVI must be enclosed by pwbm (exempt pwbm hole inside dnwell)	NV
pwbm.5	Min Space between pwbm holes inside UHVI	NV

5.7.5 pwdem Rules

Note: pwdem rules are not implemented due to lack of information.

pwdem.1	Min width of pwdem	NV
pwdem.2	Min spacing between pwdem inside UHVI on same net	NV
pwdem.3	Min enclosure of pdwem by pwbm inside UHVI	NV
pwdem.4	pwdem must be enclosed by UHVI	
pwdem.5	pwdem inside UHVI must be enclosed by deep nwell	
pwdem.6	Min enclosure of pwdem by deep nwell inside UHVI	

5.7.6 hvtp Rules

hvtp.1	CDR: Min width of hvtp	0.38 um
hvtp.2	CDR: Min space between hvtp to hvtp	0.38 um
hvtp.3	Min enclosure of pfet outside of areaid.ce by hvtp	0.18 um
hvtp.4	Min spacing between pfet and hvtp	0.18 um
hvtp.5	Min area of hvtp	0.265 um ²
hvtp.6	Min area of hvtp holes	0.265 um ²

5.7.7 hvtr Rules

hvtr.1	CDR: Min width of hvtr	0.38 um
hvtr.2	CDR: Min spacing between hvtp to hvtr	0.38 um
hvtr.3	Min enclosure of pfet by hvtr	0.18 um

5.7.8 lvtn Rules

lvtn.1a	CDR: Min width of lvtn	0.38 μm
lvtn.2	CDR: Min space lvtn to lvtn	0.38 μm
lvtn.3a	Min spacing of lvtn to gate. exempted inside UHVI:	0.18 μm
lvtn.3b	Min spacing of lvtn to pfet along the S/D direction:	0.19 μm
lvtn.4b	Min enclosure of gate by lvtn. exempted inside UHVI:	0.18 μm
lvtn.9	Min spacing, no overlap, between lvtn and hvtp:	0.38 μm
lvtn.10	Min enclosure of lvtn by (nwell not overlapping Var_channel) (exclude coincident edges)	0.38 μm
lvtn.12	Min spacing between lvtn and (nwell inside areaid.ce):	0.38 μm
lvtn.13	Min area of lvtn:	0.265 μm^2
lvtn.14	Min area of lvtn holes:	0.265 μm^2
lvtn.15	CDR: lvtn cannot straddle nwell	

5.7.9 ncm Rules

ncm.1	CDR: Min width of ncm	0.38 μm
ncm.2	CDR: Min spacing of ncm to ncm manual merge ncm if space is below minimum	0.38 μm
ncm.3	Min enclosure of P+diff by ncm	0.18 μm
ncm.4	Min enclosure of P+diff within (areaid:esd AndNot areaid:diode) by ncm	0.18 μm
ncm.5	Min space, no overlap, between ncm and (LVTN_gate) OR (diff containing lvtn)	0.23 μm
ncm.6	Min space, no overlap, between ncm and nfet:	0.20 μm
ncm.7	Min area of ncm:	0.265 μm^2
ncm.8	Min area of ncm Holes:	0.265 μm^2

5.7.10 difftap Rules

	<i>difftap width rules</i>	
difftap.1	CDR: Minimum width of diff or tap outside areaid.ce	0.15 μm
difftap.1a	CDR: Minimum width of diff or tap inside areaid.ce	0.14 μm
difftap.1b	CDR: Minimum width of diff or tap crossing areaid.ce	0.15 μm
difftap.2	Min channel width (diff and poly) for FETs outside areaid.sc & areaid.ce :	0.42 μm
difftap.2b	Min channel width (diff and poly) for FETs inside areaid.sc but outside areaid.ce :	0.36 μm
difftap.2c	Min channel width in areaid.ce:	0.14 μm
difftap.2a	Min channel width (diff and poly); TBA for cell names s8cell_ee_plus_sselp_a, s8cell_ee_plus_sselp_b, s8fpls_pl8, s8fpls_rdrv4 , s8fpls_rdrv4f and s8fpls_rdrv8	
difftap.4	Min width of tap butting diffusion	0.29 μm
difftap.5a	Min width of tap outside areaid.ce butting and between diffusion	0.4 μm
difftap.5b	Min width of tap inside areaid.ce butting and between diffusion	0.38 μm
	<i>difftap spacing rules</i>	
difftap.3	CDR: Minimum spacing of diff to diff	0.27 μm
difftap.7	Min spacing of diff/tap abutting edge to a non-coinciding diff or tap edge:	0.13 μm
difftap.9	Min spacing of (n+) diffusion to N-well outside UHVI:	0.34 μm
difftap.11	Min spacing of (p+) tap to N-well. exempted inside UHVI:	0.13 μm
	<i>difftap enclosure rules</i>	
difftap.8a	Min enclosure of (p+) diffusion by N-well; exempted inside UHVI / areaid.ed (ESD) / areaid.ce (CORE)	0.18 μm
difftap.8b	Min enclosure of (p+) diffusion in areaid.ce by N-well:	0.15 μm
difftap.8c	Min enclosure of adjacent sides of (p+) diffusion in areaid.ce by N-well:	0.18 μm

difftap.10a	Min enclosure of (n+) tap by N-well exempted inside: UHVI / areaid.ed (ESD) / areaid.ce (CORE)	0.18 μm
difftap.10b	Min enclosure of (n+) tap in areaid.ce by N-well:	0.15 μm
	difftap other rules	
difftap.6	Diff and tap are not allowed to extend beyond their abutting edge	
difftap.12	GUIDELINE: ESD_nwell_tap is considered shorted to the abutting diff <i>Rule not checked by DRC</i>	
difftap.13	Diffusion or the RF FETS in Table H5 is defined by Ldiff and Wdiff. <i>Rule not checked, informational only statement</i>	

5.7.11 tunm Rules

tunm.1	Min width of tunm:	0.41 μm
tunm.2	Min spacing of tunm to tunm:	0.50 μm
tunm.3	Extension of tunm beyond (poly and diff):	0.095 μm
tunm.4	Min spacing of tunm to (poly and diff) outside tunm:	0.095 μm
tunm.5	(poly and diff) may not straddle tunm	
tunm.6a	Tunm outside deep n-well is not allowed	
tunm.7	Min tunm area:	0.672 μm^2
tunm.8	tunm must be enclosed by areaid.ce	

5.7.12 poly Rules

poly.X.1	All FETs would be checked for W/Ls as documented in spec 001-02735. Exempt FETs that are pruned exempt for W/Ls inside areaid.sc and inside cell name scs8*decap* and listed in the MRGA as a decap only W/L <i>Rule not checked, informational only statement</i>	
poly.X.1a	Min & max dummy_poly L is equal to min L allowed for corresponding device type (exempt for dummy_poly in cells listed on Table H3) <i>Rule not checked, informational only statement</i>	
poly.1a	Min width of poly	0.15 μm
poly.1b	Min channel length (poly width) for pfet overlapping lvtm	0.35 μm
poly.2	CDR: Min spacing/notch of poly to poly outside of areaid.ce	0.21 μm
poly.2a	CDR: Min spacing of poly core gap (end-to-end spacing of poly in areaid.ce)	0.16 μm
poly.2b	CDR: Min spacing of poly inside areaid.ce (except for end-to-end spacing of poly in areaid.ce)	0.175 μm
poly.4a	Min spacing of poly on field (outside areaid.ce) to diff (parallel edges only)	0.075 μm
poly.4b	Min spacing of poly in core (inside areaid.ce) to diff	0.03 μm
poly.5a	Min spacing of poly on field (outside areaid.ce) to tap	0.055 μm
poly.5b	Min spacing of poly in core (inside areaid.ce) to tap	0.03 μm
poly.6	Min spacing of poly on diff to abutting tap (min source)	0.30 μm
poly.9	Poly resistor spacing to poly or spacing (no overlap) to diff/tap	0.48 μm
poly.9a	Min poly resistor spacing to diff/tap	0.48 μm
poly.9b	Min poly resistor spacing to poly	0.21 μm
poly.3	CDR: Min poly resistor width	0.33 μm
poly.3a	Poly resistor must be covered by poly	
poly.3b	CDR: Min length of precision poly resistor	0.66 μm
poly.3c	CDRW: Min length of non-precision poly resistor	0.47 μm
poly.7	Min extension of diff beyond poly in periphery (outside areaid.ce)	0.25 μm
poly.8	Min extension of poly outside areaid.ce beyond gate end (endcap)	0.13 μm
poly.10	Poly cannot overlap inner corners of diff	
poly.11	No 90 deg turns of poly on diff	

poly.12	(Poly NOT (nwell NOT hvi)) may not overlap tap exempted for gated_npn and inside UHVI	
poly.15	Poly must not overlap diff: rs (diff res)	
poly.16	Inside RF FETs defined in Table H5, poly cannot overlap poly across multiple adjacent instances Rule not implemented	

5.7.13 rpm / urpm Rules

rpm.1a	CDR: Min width of rpm	1.27 um
rpm.1b	Min/Max prec_resistor width xhrpoly_0p35	0.35 um
rpm.1c	Min/Max prec_resistor width xhrpoly_0p69	0.69 um
rpm.1d	Min/Max prec_resistor width xhrpoly_1p41	1.41 um
rpm.1e	Min/Max prec_resistor width xhrpoly_2p85	2.85 um
rpm.1f	Min/Max prec_resistor width xhrpoly_5p73	5.73 um
rpm.1g	Only 1 licon is allowed in xhrpoly_0p35 prec_resistor_terminal	
rpm.1h	Only 1 licon is allowed in xhrpoly_0p69 prec_resistor_terminal	
rpm.1i	Only 2 licon is allowed in xhrpoly_1p41 prec_resistor_terminal	
rpm.1j	Only 4 licon is allowed in xhrpoly_2p85 prec_resistor_terminal	
rpm.1k	Only 8 licon is allowed in xhrpoly_5p73 prec_resistor_terminal	
rpm.2	CDR: Min spacing of rpm	0.84 um
rpm.3	Min enclosure of prec_resistor by rpm	0.20 um
rpm.4	Min enclosure of prec_resistor by psdm	0.11 um
rpm.5	Min enclosure of prec_resistor by np	0.095 um
rpm.6	Min spacing, no overlap, of rpm and nsdm	0.20 um
rpm.7	Min spacing between rpm and poly	0.20 um
rpm.8	poly must not straddle rpm	
rpm.9	Min space, no overlap, between prec_resistor and hvntm	0.185 um
rpm.10	Min spacing of rpm to pwbm <i>Rule not coded, no information</i>	NV
rpm.11	rpm should not overlap or straddle pwbm	
urpm.1	CDR: Min width of urpm	1.27 um
urpm.2	CDR: Min spacing of urpm	0.84 um

5.7.14 varac Rules

varac.1	Min channel length (poly width) of Var_channel	0.18 um
varac.2	Min channel width (tap width) of Var_channel	1.00 um
varac.3	Min spacing between hvtp to Var_channel	0.18 um
varac.4	Min spacing of licon on tap to Var_channel	0.25 um
varac.5	Min enclosure of poly overlapping Var_channel by nwell	0.15 um
varac.6	Min spacing between VaracTap and difftap	0.27 um
varac.7	Nwell overlapping Var_channel must not overlap P+ diff	
varac.8	Min enclosure of Var_channel by hvtp	0.255

5.7.15 photo Rules

photo.1	Rules dnmwell.3 and nwell.5 are exempted for photoDiode	
photo.2	Min/Max width of photoDiode	3.0 um
photo.3	Min spacing between photoDiode	5.0 um
photo.4	Min spacing between photoDiode and deep nwell	
photo.5	photoDiode edges must be coincident with areaid.po	
photo.6	photoDiode must be enclosed by dnmwell ring	

photo.7	photoDiode must be enclosed by p+ tap ring	
photo.8	Min/Max width of nwell inside photoDiode	
photo.9	Min/Max enclosure of nwell by photoDiode	
photo.10	Min/Max width of tap inside photoDiode	
photo.11	Min/Max enclosure of tap by nwell inside photoDiode	

5.7.16 npc Rules

npc.1	Min width of NPC	0.27 um
npc.2	Min spacing of NPC to NPC	0.27 um
npc.3	Manual merge if less than minimum	
npc.4	Spacing (no overlap) of NPC to Gate	0.09 um
npc.5	Max enclosure of poly overlapping slotted_licon by npc; merge between adjacent short edges of the slotted_licons if space < min	0.095 um

5.7.17 n/psd Rules

psd.1a	CDR: Min width of nsdm / psdm outside areaid.ce:	0.38 um
psd.1b	CDR: Min width of nsdm / psdm inside areaid.ce:	0.29 um
psd.2	CDR: Min spacing of nsdm(psdm) to nsdm(psdm):	0.38 um
pad.3	Manual merge if less than minimum	
psd.5a	Min enclosure of diff by nsdm(psdm), except for butting edge	0.125 um
psd.5b	Min enclosure of tap by nsdm(psdm), except for butting edge	0.125 um
psd.6	Enclosure of diff/tap butting edge by nsdm (psdm)	0.0 um
psd.7	Min spacing of NSDM/PSDM to opposite implant diff or tap (for non-abutting diff/tap edges)	0.13 um
psd.8	nsdm and psdm cannot overlap diff/tap regions of opposite doping	
psd.9	WARNING: diff and tap must be enclosed by their corresponding implant layers. exempted for diff rings around the die; exempted for gated_npn & areaid.zr	
psd.10a	Min area of nsdm	0.265 um ²
psd.10b	Min area of psdm	0.255 um ²
psd.11	Min area of nsdm/psdm holes	0.265 um ²

5.7.18 licon Rules

	licon Size Rules	
licon.1	CDR: Min and max L and W of licon, except for bar licon:	0.17 um
licon.1a	licon inside (rpm OR urpm) should be 0.19 um X 2.0 um bar licon	
licon.1b	CDR: Min and max width of bar licon	0.19 um
licon.1c	CDR: Min and max length of bar licon	2.0 um
licon.3	Only min. square licons are allowed except die seal ring where licons are:	0.17 um * L
	licon Spacing Rules	
licon.2	CDR: Min spacing of licon, for non-bar licon outside of areaid.ce	0.17 um
licon.2a	CDR: Min spacing of licon, for non-bar licon inside of areaid.ce	0.165 um
licon.2b	CDR: Min end-to-end space of bar licon:	0.35 um
licon.2c	CDR: Min side-to-side spacing of bar licon:	0.51 um
licon.2d	CDR: Min space from bar licon to square licon:	0.51 um
licon.5b	CDR: Min space between tap_licon outside areaid.ce and diff-abutting tap edge	0.06 um
licon.9	CDR: Min spacing, no overlap, between poly_licon outside areaid.ce and psdm	0.11 um
licon.14	Min spacing of poly_licon outside areaid.ce to diff or tap:	0.19 um
licon.14a	Min spacing of poly_licon inside areaid.ce to diff or tap:	0.13 um
licon.2b	Min spacing between two slotted_licon (both the edges are 0.19um in length)	0.35 um
licon.2c	Min spacing between two slotted_licon (except for licon.2b)	0.51 um

licon.2d	Min spacing between a slotted_licon and 0.17um square licon	0.51 um
	Overlap Rules and SRAM Alternate Rules (see NOTE below) *	
licon.4a	licon must be inside li	
licon.4b	licon must be inside (poly or diff or tap)	
licon.4a_SRAM	*licon inside areaid.ce must overlap LI	
licon.4b_SRAM	*licon inside areaid.ce must overlap (poly or diff or tap)	
licon.WARNING	*WARNING: alternate licon.4a & licon.4b rules used in areaid.ce regions *WARNING: real licon.4a & licon.4b errors may be missed	
	licon Enclosure Rules	
licon.5a	Min enclosure of licon outside areaid.ce by diff (not tap)	
licon.5c	Min enclosure of licon outside areaid.ce by diff on one of two adjacent sides	
licon.7	Min enclosure of licon outside areaid.ce by one of two adjacent sides of isolated tap:	0.04 um
licon.8	Min enclosure of poly_licon outside areaid.ce by poly	0.06 um
licon.8a	Min enclosure of poly_licon outside areaid.ce by poly on one of two adjacent sides:	0.12 um
licon.15	poly_licon outside areaid.ce must be enclosed by npc by	0.05 um
licon.15b	CDR: poly_licon inside areaid.ce must be enclosed by npc by	0.08 um
	Other licon Rules	0.10 um
licon.6	licon cannot straddle tap	0.045 um
licon.10	Spacing of licon on (tap AND (nwell NOT hvi)) to Var_channel:	
licon.11	Spacing of licon on diff or tap to poly on diff, except for FETs inside areaid.sc	
licon.11a	CDR: Spacing of licon on diff or tap to poly on diff, for all FETs inside areaid.sc	0.25 um
licon.11b	<i>Rule is obsolete</i>	0.055 um
licon.11c	<i>Rule is obsolete</i>	0.05 um
licon.11d	<i>Rule is obsolete</i>	
licon.12	GUIDELINE: Max SD width without licon	5.7 um
licon.13	CDR: Spacing (no overlap) of NPC to licon on diff or tap	
licon.14a	Min spacing of poly_licon outside areaid.ce to diff or tap	5.7 um
licon.14b	Min spacing of poly_licon inside areaid.ce to diff or tap	0.09 um
licon.15a	Min enclosure of poly_licon outside areaid.ce by npc by	0.19 um
licon.15b	CDR: Min enclosure of poly_licon inside areaid.ce by npc	0.13 um
licon.16	Every source diff and tap must enclose at least one licon, including diff/tap straddling areaid.ce; rule exempted inside UHVI. <i>Rule not implemented</i>	0.10 um
licon.17	licon may not overlap both poly and (diff or tap)	0.045 um
licon.18	npc must enclose poly_licon	
licon.19	poly of the HV varactor must not interact with licon	
licon.20	Every tap must enclose at least 1 licon (except in uhvi)	
licon.21	psdm overlapping (poly and licon) is prohibited inside areaid.ce	
licon.core	WARNING: licon straddling areaid.ce (CORE)	

NOTE: SRAM alternate rules are enabled with environment variable / deck switch “SRAM” – this is for debugging purposes and will cause real licon.4a / licon.4b DRC violations to be missed.

5.7.19 li Rules

li.1a	CDR: Width of LI inside areaid.ce:	0.14 um
li.1b	CDR: Width of LI outside areaid.ce:	0.17 um
li.1c	CDR: Width of LI crossing areaid.ce border:	0.17 um
li.2	GUIDELINE: Max ratio of length to width of LI without licon or mcon <i>Rule not checked in DRC</i>	
li.3a	CDR: Spacing of LI to LI inside areaid.ce:	0.14 um

li.3b	CDR: Spacing of LI to LI outside areaid.ce:	0.17 um
li.3c	CDR: Spacing of LI to LI crossing areaid.ce border:	0.17 um
li.5	CDR: Min enclosure of licon outside areaid.ce by one of two adjacent LI sides:	0.08 um
li.6a	CDR: Min area of Li outside areaid.ce:	0.056 um ²
li.6b	CDR: Min area of Li crossing areaid.ce border:	0.056 um ²
li.7	CDR: Min LI resistor width 0.29 um	

5.7.20 mcon Rules

mcon.1	CDR: Min and max L and W of mcon	0.17 um
mcon.2	CDR: Min spacing of mcon to mcon	0.19 um
mcon.3	CDR: Only min. square mcons are allowed	0.17 um
mcon.4	Mcon must be enclosed by LI by at least:	0.0 um
mcon.IR.1	For 1 <= n <= 10 contacts on the same connector, mcon area pre- and post- Cu conversion must differ by no more than 0.2 um; <i>Rule not implemented</i>	
mcon.IR.2	For 11 <= n <= 100 contacts on the same connector, mcon area pre- and post- Cu conversion must differ by no more than 0.3 um; <i>Rule not implemented</i>	
mcon.IR.3	For n > 100 contacts on the same connector, mcon area pre- and post- Cu conversion must differ by no more than 0.7 um; <i>Rule not implemented</i>	
mcon.core	WARNING: mcon straddling areaid.ce (CORE)	

5.7.21 capm & cap2m Rules

capm.1a	met3 must completely enclose capm	
capm.1b	capm without met4 is prohibited	
capm.2a	capm without via3 is prohibited	
capm.2b	capm must contain at least 2 via3	
capm.3a	CDR: Min space between unconnected capm bottom plates	1.2 um
capm.3b	CDR: Min space capm bottom plates to unconnected met3	1.2 um
capm.3c	CDR: Min capm space	0.84 um
capm.4	CDR: Min capm width	1.0 um
capm.5	CDR: Min enclosure of capm by met3	0.14 um
capm.6	CDR: Min enclosure of via3 by capm	0.14 um
capm.7	CDR: Min spacing of capm and via3	0.14 um
capm.8a	Max capm aspect ratio (L/W)	20.0 um
capm.8b	Only rectangular capm is permitted	
capm.8c	Max area of capm	1,000,000 um ²
capm.9a	Min space of capm to via2	0.14 um
capm.9b	capm cannot overlap via2	
cap2m.1a	met4 must completely enclose cap2m	
cap2m.1b	cap2m without met5 is prohibited	
cap2m.2a	cap2m without via4 is prohibited	
cap2m.2b	cap2m must contain at least 2 via4	
cap2m.3a	CDR: Min space between unconnected cap2m bottom plates	1.2 um
cap2m.3b	CDR: Min space cap2m bottom plates to unconnected met4	1.2 um
cap2m.3c	CDR: Min cap2m space	0.84 um
cap2m.4	CDR: Min cap2m width	1.0 um
cap2m.5	CDR: Min enclosure of cap2m by met4	0.14 um
cap2m.6	CDR: Min enclosure of via4 by cap2m	0.20 um
cap2m.7	CDR: Min spacing of cap2m and via4	0.20 um
cap2m.8a	Max cap2m aspect ratio (L/W)	20.0 um
cap2m.8b	Only rectangular cap2m is permitted	

cap2m.8c	Max area of ca2pm	1,000,000 μm^2
cap2m.9a	Min space of cap2m to via3	0.14 μm
cap2m.9b	cap2m cannot overlap via3 except for stacked capm / cap2m capacitors	

5.7.22 vpp Rules

VPP capacitors are no longer supported by SkyWater Technology. Basic width & spacing rules are retained as GUIDELINE rules for legacy devices; other rules are now obsolete.

vpp.1	GUIDELINE: Min width of capacitor.dg	1.43 μm
vpp.1b	<i>Rule is obsolete</i>	
vpp.1c	<i>Rule is obsolete</i>	
vpp.3	<i>Rule is obsolete</i>	
vpp.4	<i>Rule is obsolete</i>	
vpp.5	<i>Rule is obsolete</i>	
vpp.8	GUIDELINE: Min enclosure of capacitor.dg by nwell	1.5 μm
vpp.9	GUIDELINE: Min spacing of capacitor.dg to nwell	1.5 μm
vpp.10	<i>Rule is obsolete</i>	
vpp.12a	<i>Rule is obsolete</i>	
vpp.12b	<i>Rule is obsolete</i>	
vpp.12c	<i>Rule is obsolete</i>	
vpp.13	GUIDELINE: Min space of met1 to met1 inside VPP capacitor	0.16 μm
vpp.14	GUIDELINE: Min space of met2 to met2 inside VPP capacitor	0.16 μm

5.7.23 met1 Rules

m1.1	CDR: Min width of met1	0.14 μm
m1.2	CDR: Min spacing/notch of met1	0.14 μm
m1.3a	CDRW: Min spacing of huge met1 (width > 3.0 μm) to narrow met1, including features extending 0.28 μm into narrow met1	0.28 μm
m1.3b	CDRW: Min spacing of huge_met1 to met1 excluding features checked by m1.3a	0.28 μm
m1.4	CDR: Min enclosure of mcon outside areaid.ce by met1	0.03 μm
m1.4a	CDR: Min enclosure of mcon inside areaid.ce by met1	0.005 μm
m1.5	CDR: Min enclosure of mcon outside areaid.ce by met1 on one of two adjacent sides	0.06 μm
m1.6	CDR: Min met1 area	0.083 μm^2
m1.7	CDR: Min area of met1 holes:	0.14 μm^2
m1.pd.1 *	*Rule is renamed "m1.local.min" and moved to the Density Rules section	80%
	<i>Cu Rules included for reference</i>	
m1.11	Cu Flow - GUIDELINE: Max width of met1 after slotting	4.0 μm
m1.12	Cu Flow - Add slots and remove vias & contacts if met1 wider than:	3.20 μm
m1.15	Cu Flow - Min enclosure of mcon by met1 on one of two adjacent sides	0.03 μm
m1.13	Cu Flow - Max pattern density of met1	77%
m1.14	met1 PD window size 50 μm	
m1.14a	met1 PD window step 25 μm	

Note: The following met1 Recommended Chip-Level Density Rules are obsolete and removed:

m1.RC.pd1 / m1.RC.pd2 / m1.RC.pd3 / m1.RC.pd4 / m1.RC.pd5 / m1.RC.pd6

5.7.24 via1 Rules

via1.3	Only minimum square vias are allowed except die seal ring where vias are:	0.20 μm * L
via1.1a	CDR: Min and max L and W of via1 outside areaid.mt	0.15 μm
via1.1b	Three sizes of square via1's allowed inside areaid.mt:	

	0.15 um, 0.23 um and 0.28 um	
via1.2	CDR: Min spacing of via1	0.17 um
via1.4a	CDR: via1 outside of areaid.mt must be enclosed by met1 by at least	0.055 um
via1.4b	Inside areaid.mt, 0.23 um via1 must be enclosed by met1 by at least	0.03 um
via1.4c	Inside areaid.mt, 0.28 um via1 must be enclosed by met1 by at least	0. um
via1.5a	CDR: Min enclosure of 0.15 um via1 by met1 on one of two adjacent sides	0.085 um
via1.5b	Min enclosure of 0.23 um via1 by met1 on one of two adjacent sides	0.085 um
via1.5c	Inside areaid.mt, min enc. of 0.28 um via1 by met1 on one of two adjacent sides	0.00 um
	<i>Cu Rules included for reference</i>	
via1.11	Cu FLOW - Min and max L and W of via1 outside areaid.mt:	0.18 um
via1.12	Cu Flow – Min spacing of via1 to via1:	0.13 um
via1.13	Cu Flow - Max of 5 vias within 0.35 um	
via1.14	Cu Flow – Min enclosure of 0.18 um via1 by parallel edges of met1	0.04 um
via1.irdrc.1	<i>IR rule, not implemented in DRC</i>	
via1.irdrc.2	<i>IR rule, not implemented in DRC</i>	
via1.irdrc.3	<i>IR rule, not implemented in DRC</i>	
via1.irdrc.4	<i>IR rule, not implemented in DRC</i>	

5.7.25 m2 Rules

m2.1	CDR: Min width of met2	0.14 um
m2.2	CDR: Min spacing/notch of met2	0.14 um
m2.3a	CDRW: Min spacing of huge met2 (width > 3.0 um) to narrow met2, including features extending 0.28 um into narrow met2	0.28 um
m2.3b	CDRW: Min spacing of huge_met2 to met2 excluding features checked by m2.3a	0.28 um
m2.3c	<i>Rule is obsolete</i>	
m2.4a	CDR: Min enclosure of via1 outside areaid.ce by met2	0.055 um
m2.4b	CDR: Min enclosure of via1 inside areaid.ce by met2 by at least	0.045 um
m2.5	CDR: Min enclosure of via1 outside areaid.ce by met2 on one of two adjacent sides	0.085 um
m2.6	Min met2 area	0.067 um ²
m2.7	Min area of met2 holes	0.14 um ²
m2.pd.1*	*Rule is renamed “m2.local.min” and moved to the Density Rules section	80%
	<i>Cu Rules included for reference</i>	
m2.11	Cu Flow - Max width of met2:	4.0 um
m2.12	Cu Flow - Add slots and remove vias and contacts if met2 wider than:	3.2 um
m2.15	Cu Flow - via1 must be enclosed by met2 on one of two adjacent sides by at least:	0.04 um
m2.13	Cu Flow - max pattern density (PD) of met2	77%
m2.14	met2 PD window size 50 um	
m2.14a	met2 PD window step 25 um	

Note: The following met2 Recommended Chip-Level Density Rules are obsolete and removed:
m2.RC.pd1 / m2.RC.pd2 / m2.RC.pd3 / m2.RC.pd4 / m2.RC.pd5 / m2.RC.pd6

5.7.26 via2 Rules

via2.1a	CDR: Min and max L and W of via2 outside areaid.mt	0.20 um
via2.1b	<i>Rule has conflicting information and is removed</i>	
via2.1c	<i>Rule has conflicting information and is removed</i>	
via2.1e	<i>Rule has conflicting information and is removed</i>	
via2.1f	<i>Rule has conflicting information and is removed</i>	
via2.1d	AL Flow - Four sizes of square Via2s allowed inside areaid.mt: 0.20 um, 0.28 um, 1.20 um and 1.50 um	

via2.2	CDR: Min spacing of via2	0.20 um
via2.3	Only min. square via2s are allowed except die seal ring where vias are:	0.20 um * L
via2.4	CDR: Min enclosure of via2 by met2	0.04 um
via2.4a	Inside areaid.mt, min enclosure of 1.5 um via2 by met2	0.14 um
via2.5	CDR: Min enclosure of via2 by met2 on one of two adjacent sides	0.085 um
via2.11	Cu FLOW - Min and max L and W of via2 == 0.21 um	
via2.12	Cu Flow - Spacing of via2 to via2:	0.18 um
via2.13	Cu Flow - Min Spacing between via2 rows:	0.20 um
via2.14	Cu Flow - Min enclosure of via2 by met2:	0.035 um
via2.irdrc.1	<i>IR rule, not implemented in DRC</i>	
via2.irdrc.2	<i>IR rule, not implemented in DRC</i>	
via2.irdrc.3	<i>IR rule, not implemented in DRC</i>	

5.7.27 m3 Rules

m3.1	CDR: Min width of met3	0.30 um
m3.2	CDR: Min spacing/notch of met3	0.30 um
m3.3a	Min spacing of features attached to or extending from huge_met3 for a distance of up to 0.48 um to met3 (not checked over non-huge met3 features)	NV
m3.3b	Min spacing of huge_met3 to met3 excluding features checked by m3.3a	NV
m3.3c	CDRW: Min spacing of huge met3 (width > 3.0 um) to narrow met3, including features extending 0.28 um into narrow met3	0.40 um
m3.3d	CDRW: Min spacing of huge_met3 to met3 excluding features checked by m3.3c	0.40 um
m3.4	Min enclosure of via2 by met3 by at least	0.065 um
m3.5	Min enclosure of via2 by met3 on one of two adjacent sides	NV
m3.5a	Min enclosure of via2 by met3 on all sides (not checked on a layout when it satisfies both Rules m3.4 and m3.5)	NV
m3.6	CDR: Min met3 area	0.240 um ²
m3.7	Min area of met3 holes	0.20 um ²
m3.pd.1*	*Rule is renamed “m3.local.min” and moved to the Density Rules section	80%
	<i>Cu Rules included for reference</i>	
m3.11	Cu Flow - Max width of met3	4.0 um
m3.12	Cu Flow - Add slots and remove vias and contacts if met3 wider than:	3.2 um
m3.15	Cu Flow - Min enclosure of via2 by met3 on one of two adjacent sides	0.06 um
m3.13	Cu Flow - Max pattern density (PD) of met3 (CU)	77%
m3.14	met3 PD window size 50 um	
m3.14a	met3 PD window step 25 um	

Note: The following met3 Recommended Chip-Level Density Rules are obsolete and removed:

m3.RC.pd1 / m3.RC.pd2 / m3.RC.pd3 / m3.RC.pd4 / m3.RC.pd5 / m3.RC.pd6

5.7.28 via3 Rules

via3.1	CDR: Min and max L and W of via3 (except for via3.1a)	0.20 um
via3.1a	Two sizes of square via3 allowed inside areaid.mt: 0.20 um & 0.80 um	
via3.2	CDR: Spacing of via3 to via3	0.20 um
via3.3	Only min. square via3s are allowed except die seal ring where via3s are:	0.20 * L
via3.4	CDR: Min enclosure of via3 by met3	0.06 um
via3.5	CDR: Min enclosure of via3 by met3 on one of two adjacent sides	0.09 um
via3.11	Cu FLOW - Min and max L and W of via3	0.21 um
via3.12	Cu Flow - Min spacing of via3 to via3	0.18 um
via3.13	Cu Flow - Min enclosure of via3 by met3	0.055 um

via3.14	Cu Flow - Min Spacing between via3 rows	0.35 um
via3.irdrc.1	<i>IR rule, not implemented in DRC</i>	
via3.irdrc.2	<i>IR rule, not implemented in DRC</i>	
via3.irdrc.3	<i>IR rule, not implemented in DRC</i>	
via3.irdrc.4	<i>IR rule, not implemented in DRC</i>	

5.7.29 nsm Rules

nsm.1	CDR: Min width of nsm	3.0 um
nsm.2	CDR: Min spacing of nsm	4.0 um
nsm.3	Min spacing, no overlap, between NSM_keepout to diff.dg, tap.dg, fom.dy, cfom.dg, cfom.mk, poly.dg, p1m.mk, li1.dg, cli1m.mk, metX.dg (X=1 to 5) and cmmX.mk (X=1 to 5). Exempt: (a) cell name nikon and (b) diff ring inside areaid.sl:	1.0 um
nsm.3a	Min enclosure of diff.dg, tap.dg, fom.dy, cfom.dg, cfom.mk, poly.dg, p1m.mk, li1.dg, cli1m.mk, metX.dg (X=1 to 5) and cmmX.mk (X=1 to 5) by areaid.ft. Exempt the following from the check: (a) cell name s8Fab_crntic (b) blankings in the frame (areaid.dt)	3.0 um
nsm.3b	Min spacing between areaid.dt to diff.dg, tap.dg, fom.dy, cfom.dg, cfom.mk, poly.dg, p1m.mk, li1.dg, cli1m.mk, metX.dg (X=1 to 5) and cmmX.mk (X=1 to 5). Exempt the following from the check: (a) blankings in the frame (areaid.dt)	3.0 um

5.7.30 indm Rules

No inductor rules are coded as no information is available; rule descriptions shown for reference.

indm.1	Min width of top inductor metal	NV
indm.2	Min spacing between top inductor metal	NV
indm.3	Min enclosure of via by top inductor metal	NV
indm.4	Min area of top inductor metal	NV

5.7.31 m4 Rules

m4.1	CDR: Min width of met4	0.30 um
m4.2	CDR: Min spacing/notch of met4	0.30 um
m4.3	CDR: Min enclosure of via3 by met4	0.065 um
m4.4a	CDR: Min met4 area	0.240 um ²
m4.5a	CDRW: Min spacing of huge met4 (width > 3.0 um) to narrow met4, including features extending 0.28 um into narrow met4	0.40 um
m4.5b	CDRW: Min spacing of huge_met4 to met4 excluding features checked by m4.3c	0.40 um
m4.pd.1*	*Rule is renamed "m4.local.min" and moved to the Density Rules section	80%
	<i>Cu Rules included for reference</i>	
m4.7	Cu Flow - Min area of met4 holes	0.200 um ²
m4.11	CU Flow - Max width of met4	10.0 um
m4.12	CU Flow - Add slots and remove vias and contacts if met4 wider than:	10.0 um
m4.15	CU Flow - Min enclosure of via3 by met4	0.06 um
m4.16	CU Flow - Min enclosure of pad by met4	0.85 um
m4.13	CU Flow - Max pattern density (PD) of met4	77%
m4.14	window size 5 um	
m4.14a	window step 25 um	

5.7.32 via4 Rules

via4.1	CDR: Min and max L and W of via4	0.80 um
via4.2	CDR: Min spacing of via4:	0.80 um
via4.3	Only min. square via4s are allowed except die seal ring where via4s are"	0.80 * L
via4.4	CDR: Min enclosure of via4 by met4	0.19 um

via4.irdrc.1	<i>IR rule, not implemented in DRC</i>	
via4.irdrc.2	<i>IR rule, not implemented in DRC</i>	
via4.irdrc.3	<i>IR rule, not implemented in DRC</i>	
via4.irdrc.4	<i>IR rule, not implemented in DRC</i>	

5.7.33 m5 Rules

m5.1	CDR: Width of met5	1.60 μm
m5.2	CDR: Min spacing/notch of met5	1.60 μm
m5.3	CDR: Min enclosure of via4 by met5	0.31 μm
m5.4	CDR: Min area of met5	4.0 μm^2

5.7.34 pad Rules

pad.2	Min spacing of pad	1.27 μm
pad.3	Min area of hugePad NOT top_metal	30,000 μm^2

5.7.35 RDL Rules

RDL is not supported for this process.

5.7.36 mf Rules

Fuse devices are not supported for this process.

5.7.37 hvi Rules

hvi.nwell	hvi must not straddle nwell	
hvi.1	Min width of hvi	0.02 μm
hvi.5	Min space between hvi and nwell; hvi.5 is obsolete due to additions of nwell spacing rules nwell.2a.XXX & nwell.8.XXX	0.70 μm
hvi.2	Min spacing of hvi to hvi; manual merge if space is below minimum	0.70 μm
hvi.4	hvi must not overlap tunm	

5.7.38 hvnwell Rules

hvnwell.8	Min space between HV nwell and any nwell on different nets <i>Rule is obsolete and replaced with hvnwell.8.## rules</i>	2.00 μm
hvnwell.8.hv	Min space/notch of HV_nwell	1.27 μm
hvnwell.8.hv_vhv	Min space of HV_nwell to VHV_nwell	2.0 μm
hvnwell.8.hv_uhv	Min space of HV_nwell to UHV_nwell	2.5 μm
hvnwell.8.vhv	Min space/notch of VHV_nwell	2.0 μm
hvnwell.8.vhv_uhv	Min space of VHV_nwell to UHV_nwell	2.5 μm
hvnwell.8.uvh	Min space/notch of UHV_nwell	2.5 μm
hvnwell.9	(Nwell overlapping hvi) must be enclosed by hvi	
hvnwell.10	LVnwell and HVnwell should not be on the same net (for the purposes of this check, short the connectivity through resistors) Exempt HVnwell with li nets tagged lv_net using text.dg and HVnwell connected to nwell overlapping areaid.hl <i>Rule not implemented</i>	
hvnwell.11	<i>Rule is obsolete</i>	

5.7.39 hvdifftap Rules

hvdifftap.14	Min width of diff (not tap) inside hvi, except HV pdiff resistors (hvdifftap.14a)	0.29 μm
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hvdifftap.14a	Min width of diff inside hvi, HV pdiff resistors only	0.15 μm
hvdifftap.15a	Min space of diff inside hvi	0.30 μm
hvdifftap.15b	Min space of ndiff to non-abutting ptap inside hvi:	0.37 μm
hvdifftap.16	Min width tap butting diff on one or two sides inside hvi; exempted inside UHVI	0.70 μm
hvdifftap.17	Min enclosure of HV pdiff or pdiff inside areaid:hvnwell by HV nwell; exempted inside UHVI	0.33 μm
hvdifftap.18	Min spacing of ndiff to HV nwell; exempted inside UHVI	0.43 μm
hvdifftap.19	Min enclosure of HV ntap by HV nwell; exempted inside UHVI	0.33 μm
hvdifftap.20	Min spacing of ptap to HV nwell exempted for ptap butting pwell.rs, exempted inside UHVI	0.43 μm
hvdifftap.21	Diff or tap cannot straddle hvi	
hvdifftap.22	Min enclosure of HV diff or HV tap by hvi; exempted inside UHVI	0.18 μm
hvdifftap.23	Min space between diff or tap outside hvi and hvi <i>This rule is obsolete and removed</i>	0.18 μm
hvdifftap.24	Min spacing of nwell to HV ndiff (exempted inside UHVI)	0.43 μm
hvdifftap.25	CDR: Min space of HV ndiff inside hvi across non-abutting ptap	1.07 μm
hvdifftsp.26	Min spacing between pwbm to difftap outside UHVI; Not Coded in DRM	

5.7.40 hvpoly Rules

hvpoly.13	Min width of poly over diff inside hvi:	0.50 μm
hvpoly.14	(poly and diff) cannot straddle hvi	

5.7.41 hvntm Rules

hvntm.X.1	hvntm can be drawn inside hvi.	
hvntm.1	Min width of hvntm	0.70 μm
hvntm.2	Min spacing of hvntm	0.70 μm
hvntm.3	Min enclosure of (ndiff inside hvi) but not overlapping areaid.ce by hvntm:	0.185 μm
hvntm.4	Min space between (ndiff outside Hvi) and hvntm; no overlap allowed	0.185 μm
hvntm.5	Min space between pdiff and hvntm; no overlap allowed	0.185 μm
hvntm.6a	Min space between ptap and hvntm (except along the diff-butting edge) No overlap allowed	0.185 μm
hvntm.6b	Min space between ptap and hvntm along the diff-butting edge No overlap allowed	0.0 μm
hvntm.7	Min enclosure of (ESD_nwell_tap inside hvi) by hvntm; no value provided	NV
hvntm.9	hvntm must not overlap areaid.ce	
hvntm.10	hvntm must overlap hvi	

5.7.42 denmos Rules

denmos.1	Min width of de_nFet_gate:	1.055 μm
denmos.2	Min width of de_nFet_source not overlapping poly:	0.28 μm
denmos.3	Min width of de_nFet_source overlapping poly:	0.925 μm
denmos.4	Min width of the de_nFet_drain:	0.17 μm
denmos.5	Min/Max extension of de_nFet_source over nwell	0.225 μm
denmos.6	Min/Max spacing between de_nFet_drain and de_nFet_source	1.585 μm
denmos.7	Min channel width for de_nFet_gate:	5.0 μm
denmos.8	90 degree angles are not permitted for nwell overlapping de_nFET_drain	
denmos.9a	GUIDELINE: All bevels on nwell are 45 degree, 0.43 μm from corners	
denmos.9b	GUIDELINE: All bevels on de_nFet_drain are 45 degree, 0.05 μm from corners	
denmos.10	Min enclosure of de_nFet_drain by nwell:	0.66 μm
denmos.11	Min spacing between p+ tap and (nwell overlapping de_nFet_drain):	0.86 μm

denmos.12	Min spacing between nwells overlapping de_nFET_drain:	2.40 um
denmos.13	de_nFet_source must be enclosed by nsdm by:	0.13 um
denmos.14	<i>Rule is obsolete</i>	

5.7.43 depmos Rules

depmos.1	Min width of de_pFet_gate	1.05 um
depmos.2	Min width of de_pFet_source not overlapping poly	0.28 um
depmos.3	Min width of de_pFet_source overlapping poly	0.92 um
depmos.4	Min width of the de_pFet_drain	0.17 um
depmos.5	Min/Max extension of de_pFet_source beyond nwell	0.26 um
depmos.6	Min/Max spacing between de_pFet_drain and de_pFet_source	1.19 um
depmos.7	Min channel width for de_pFet_gate	5.0 um
depmos.8	90 degree angles are not permitted for nwell overlapping de_pFET_drain	
depmos.9a	GUIDELINE: All bevels on nwell are 45 degree, 0.43 um from corners	
depmos.9b	GUIDELINE: All bevels on de_nFet_drain are 45 degree, 0.05 um from corners	
depmos.10	Min enclosure of de_pFet_drain by nwell hole	0.86 um
depmos.11	Min spacing between n+ tap and (nwell hole enclosing de_pFet_drain)	0.66 um
depmos.12	Min enclosure of de_pFet_source must by psdm	0.13 um
depmos.13	<i>Rule is obsolete</i>	

5.7.44 extd Rules

Not coded, no values provided

5.7.45 hv Rules

hv.X.1	High voltage source/drain regions must be tagged by diff: hv	
hv.X.3	High voltage poly can be drawn over multiple diff regions that are ALL reverse-biased by at least 300 mV (existence of reverse-bias is not checked by the CAD flow). It can also be drawn over multiple diffs when all sources and all drain are shorted together. In these case, the high voltage poly can be tagged with the text: dg label with a value 'hv_bb'. Exceptions to this use of the hv_bb label must be approved by SkyWater Technology. Under certain bias conditions, high voltage poly tagged with hv_bb can cross an nwell boundary. The poly of the drain extended device crosses nwell by construction and can be tagged with the 'hv_bb' label. Use of the hv_bb label on high voltage poly crossing an nwell boundary must be approved by SkyWater Technology. All high voltage poly tagged with hv_bb will not be checked to hv.poly.1, hv.poly.2, hv.poly.3 and hv.poly.4.	
hv.X.4	Any piece of layout that is shorted to hv_source/drain becomes a high voltage feature.	
	Remaining HV rules are not coded; provided information is out-of-date.	

5.7.46 vhvi Rules

vhvi.uhvi	vhvi must not overlap uhvi	
vhvi.vhv.1	GUIDELINE: Terminals operating at nominal 12V (maximum 16V) bias must be tagged as Very-High-Voltage (VHV) using vhvi: dg layer	
vhvi.vhv.2	GUIDELINE: A source or drain of a drain-extended device can be tagged by vhvi: dg. A device with either source or drain (not both) tagged with vhvi: dg serves as a VHV propagation stopper	
vhvi.vhv.3	GUIDELINE: Any feature connected to VHVSOURCE/Drain becomes a very-high-voltage feature	

vhvi.vhv.4	GUIDELINE: Any feature connected to VHVPoly becomes a very-high-voltage feature	
vhvi.vhv.5	GUIDELINE: Diffusion that is not a part of a drain-extended device (i.e., diff not areaid:en) must not be on the same net as VHVourceDrain. Only diffusion inside areaid.ed and LV diffusion tagged with vhvi: dg are exempted.	
vhvi.vhv.6	GUIDELINE: Poly resistor can act as a VHVPoly propagation stopper. For this, it should be tagged with text 'vhv_block'	
vhvi.1	Min width of vhvi:dg	0.02 um
vhvi.2	vhvi:dg cannot overlap areaid:ce	
vhvi.3	VHVGate must overlap hvi:dg	
vhvi.4	Poly connected to the same net as a VHVSOURCEDrain must be tagged with vhvi:dg layer; Rule not implemented	
vhvi.5	Vhvi:dg cannot straddle VHVSOURCEDrain	
vhvi.6	Vhvi:dg overlapping VHVSOURCEDrain must not overlap poly	
vhvi.7	Vhvi:dg cannot straddle VHVPoly	
vhvi.8	Min space between nwell tagged with vhvi:dg and deep nwell, nwell, or ndiff on a separate net (except for ndiff overlapping nwell tagged with vhvi:dg):	11.24 um

5.7.47 uhvi Rules

uhvi.1	diff/tap cannot straddle UHVI	
uhvi.2	poly cannot straddle UHVI	
uhvi.3	pwbm must be enclosed by UHVI (exempt inside areaid:lvnet)	
uhvi.4	dnwell cannot straddle UHVI	
uhvi.5	<i>Rule is obsolete</i>	
uhvi.6	UHVI must enclose dnwell	
uhvi.7	natfet.dg must be enclosed by UHVI layer by at least N/A	
uhvi.8	Minimum width of natfet.dg:	NV
uhvi.9	Minimum spacing of natfet.dg:	NV

5.7.48 ulvt Rules

Not coded, no values in DRM

5.7.49 pwres Rules

pwres.1	GUIDELINE: Pwell resistor has to be enclosed by the res layer	
pwres.2	Min/Max width of pwell resistor	2.65 um
pwres.3	Min length of pwell resistor	26.50 um
pwres.4	Max length of pwell resistor	265.0 um
pwres.5	Min/Max spacing of tap inside the pwell resistor to nwell	0.22 um
pwres.6	Min/Max width of tap inside the pwell resistor	0.53 um
pwres.7a	Every pwres_terminal must enclose 12 licon	
pwres.7b	Every pwres_terminal must enclose 12 mcons if routed through met1	
pwres.8	Diff or poly is not allowed in the pwell resistor	
pwres.9	Nwell surrounding the pwell resistor must have a full ring of contacted tap strapped with metal	
pwres.10	The res layer must abut pwres_terminal on opposite and parallel edges	
pwres.11	The res layer must abut nwell on opposite and parallel edges not checked in pwres.10	

Density Rules

Rule	Description	Value
diff.local.low	CDR: Min diff local density (700 square um window, 70 um step)	28%
diff.local.high	CDRW: Max diff local density (700 square um window, 70 um step)	62%
poly.local.low	CDRW: Min poly local density (700 square um window, 70 um step)	8%
poly.local.high	CDRW: Max poly local density (700 square um window, 70 um step)	38%
li.global.min	CDRW: Min global density of LI inside seal ring	33%
li.global.max	CDRW: Max global density of LI inside seal ring	62%
m1.global.min	CDRW: Min global density of met1 inside seal ring	33%
m1.global.max	CDRW: Max global density of met1 inside seal ring	62%
m1.local.min	CDR: Min met1 oxide pattern local density (700 square um window, 70 um step) “met1 oxide pattern” is all met5 sized by 0.6 um	80%
m2.global.min	CDRW: Min global density of met2 inside seal ring	33%
m2.global.max	CDRW: Max global density of met2 inside seal ring	62%
m2.local.min	CDR: Min met2 oxide pattern local density (700 square um window, 70 um step) “met2 oxide pattern” is all met5 sized by 0.6 um	80%
m3.global.min	CDRW: Min global density of met3 inside seal ring	33%
m3.global.max	CDRW: Max global density of met3 inside seal ring	62%
m3.local.min	CDR: Min met3 oxide pattern local density (700 square um window, 70 um step) “met3 oxide pattern” is all met5 sized by 1.15 um	80%
m4.global.min	CDRW: Min global density of met4 inside seal ring	33%
m4.global.max	CDRW: Max global density of met4 inside seal ring	62%
m4.local.min	CDR: Min met4 oxide pattern local density (700 square um window, 70 um step) “met4 oxide pattern” is all met5 sized by 1.15 um	80%
m5.global.min	CDRW: Min global density of met5 inside seal ring	43%
m5.global.max	CDRW: Max global density of met5 inside seal ring	78%
m5.local.min	Min met5 oxide pattern local density (700 square um window, 70 um step) “met5 oxide pattern” is all met5 sized by 1.15 um NOTE: rule is valid only if “viatop” is present; viatop not currently offered	80%

Other Rules

This section has rules that apply to multiple layers.

Note on Coverage Rules:

licon inside LI and (poly or diff or tap) covered by rules licon.4a & licon.4b

poly licon inside npc is covered by licon.18

	<i>Contact and Via coverage rules</i>	
mcon.cover.1	CDR: mcon must be inside li	
mcon.cover.1_SRAM	mcon inside areaid.ce must overlap li	
mcon.WARNING	WARNING: alternate mcon.cover.1 used in areaid.ce regions WARNING: real mcon.cover.1 errors may be missed	
mcon.cover.2	mcon must be inside met1	
via1.cover.1	via1 must be inside met1	
via1.cover.2	via1 must be inside met2	
via2.cover.1	via2 must be inside met2	
via2.cover.2	via2 must be inside met3	
via3.cover.1	via3 must be inside met3	
via3.cover.2	via3 must be inside met4	
via4.cover.1	via4 must be inside met4	
via4.cover.2	via4 must be inside met5	

	<i>Resistor Layer Rules</i>	
diff.tap.res	WARNING: diff res must fit exactly inside diff drawing	
li.res	WARNING: li res must fit exactly inside li drawing	
m1.res	WARNING: met1 res must fit exactly inside met1 drawing	
m2.res	WARNING: met2 res must fit exactly inside met2 drawing	
m3.res	WARNING: met3 res must fit exactly inside met3 drawing	
m4.res	WARNING: met4 res must fit exactly inside met4 drawing	
m5.res	WARNING: met5 res must fit exactly inside met5 drawing	
	<i>Blockage Layer Rules</i>	
li.block.1	li cannot overlap li blockage	
m1.block.1	met1 cannot overlap met1 blockage	
m2.block.1	met2 cannot overlap met2 blockage	
m3.block.1	met3 cannot overlap met3 blockage	
m4.block.1	met4 cannot overlap met4 blockage	
m5.block.1	met5 cannot overlap met5 blockage	
li.block.2	Min spacing of li to li blockage:	0.17 um
m1.block.2	Min spacing of met1 to met1 blockage:	0.14 um
m2.block.2	Min spacing of met2 to met2 blockage:	0.14 um
m3.block.2	Min spacing of met3 to met3 blockage:	0.3 um
m4.block.2	Min spacing of met4 to met4 blockage:	0.3 um
m5.block.2	Min spacing of met5 to met5 blockage:	1.6 um
	<i>Rules for Obsolete Fill Layers</i>	
diff.oldfill.1	WARNING: cfom/fill (GDS 22:99) is not supported. Use diff/fill (GDS 65:99)	
diff.oldfill.2	WARNING: cfom/waffleDrop (GDS 22:24) is not supported. Use diff/fill_block (GDS 65:98)	
poly.oldfill.1	WARNING: cp1m/fill (GDS 28:28) is not supported. Use poly/fill (GDS 66: 99)	
poly.oldfill.2	WARNING: cp1m/waffleDrop (GDS 33:24) is not supported. Use poly/fill_block (GDS 66: 98)	
li.oldfill.1	WARNING: cli1m/fill (GDS 56:28) is not supported. Use li/fill (GDS 67:99)	
m1.oldfill.1	WARNING: cmm1/fill (GDS 36:28) is not supported. Use met1/fill (GDS 68:99)	
m1.oldfill.2	WARNING: cmm1/waffleDrop (GDS 62:24) is not supported. Use met1/fill_block (GDS 68:98)	
m2.oldfill.1	WARNING: cmm2/fill (GDS 41:28) is not supported. Use met2/fill (GDS 69:99)	
m2.oldfill.2	WARNING: cmm2/waffleDrop (GDS 105:52) is not supported. Use met2/fill_block (GDS 69:98)	
m3.oldfill.1	WARNING: cmm3/fill (GDS 34:28) is not supported. Use met3/fill (GDS 70:99)	
m3.oldfill.2	WARNING: cmm3/waffleDrop (GDS 107:52) is not supported. Use met3/fill_block (GDS 70:98)	
m4.oldfill.1	WARNING: cmm4/fill (GDS 51:28) is not supported. Use met4/fill (GDS 71:99)	
m4.oldfill.2	WARNING: cmm4/waffleDrop (GDS 112:4) is not supported. Use met4/fill_block (GDS 71:98)	
m5.oldfill.1	WARNING: cmm5/fill (GDS 59:28) is not supported. Use met5/fill (GDS 72:99)	
m5.oldfill.2	WARNING: cmm5/waffleDrop (GDS 117:4) is not supported. Use met5/fill_block (GDS 72:98)	